

1. Administrative & Study Identification

Full Study Title: A Modular Phase I, Open-Label, Multicentre Study to Assess the Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumour Activity of Ceralasertib in Combination With Cytotoxic Chemotherapy and/or DNA Damage Repair/Novel Anti-cancer Agents in Patients With Advanced Solid Malignancies **Short Title/Acronym:** Ascending Doses of Ceralasertib in Combination With Chemotherapy and/or Novel Anti Cancer Agents **Protocol Number:** D5330C00004 **NCT Number:** NCT02264678 **Posting ID:** 415539917422220602 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Ceralasertib (AZD6738) **Phase of Development:** Phase 1 / Phase 2 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the safety and tolerability of ceralasertib administered orally in combination with cytotoxic chemotherapy regimens and/or novel anti-cancer agents, and to define the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). Also, to investigate the effect of food on ceralasertib absorption and the effect of ceralasertib on ECG parameters. * **Secondary Objectives:** To evaluate preliminary anti-tumour activity (including Objective Response Rate [ORR] and Progression-Free Survival [PFS]), pharmacokinetics (PK), and pharmacodynamics (PD) of the combination therapies.

2.2 Background & Rationale To evaluate the therapeutic potential of targeting the DNA damage response (DDR) network. Ataxia telangiectasia and Rad3-related (ATR) kinase is critical in repairing DNA damage. Preclinical models suggest that combining the ATR inhibitor ceralasertib with other agents (such as the PARP inhibitor olaparib, durvalumab, AZD5305, or carboplatin) can overcome acquired resistance (e.g., PARP resistance) and enhance anti-tumor efficacy in patients with advanced solid malignancies (including ATM-deficient or HRR-deficient tumors).

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional (Modular dose-escalation [Part A] and dose-expansion [Part B]) * **Control Type:** Single-arm (Non-randomized, Parallel Assignment across modules) * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * Target N\$: 357 enrolled participants * **Number of Sites:** Global trial (including centers in North America, Europe, and South Korea) * **Estimated Study Start:** 31-Oct-2014 * **Estimated Primary Completion:** 04-Mar-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Aged ≥ 18 years. 2. Histologic or cytologic confirmation of an advanced solid malignant tumour that is not considered appropriate for further standard treatment. 3. **Module 2 Part B expansions & Module 3:** Patients must have measurable disease with a tumour at least 1 cm in size that can be measured using a CT or MRI scan. 4. **Module 2 Part B (except B5):** No previous treatment with a PARP inhibitor. 5. **Module 2 Part B1:** Advanced gastric adenocarcinoma (including GEJ) patients with ATM deficient tumours. 6. **Module 2 Part B2:** Advanced gastric adenocarcinoma (including GEJ) patients with ATM proficient tumours. 7. **Module 2 Part B3:** Second or third line HER2 negative breast cancer. 8. **Module 2 Part B4:** Second or third line triple negative breast cancer (TNBC). 9. **Module 2 Part B5:** BRCAm or RAD51C/Dm or PALB2m or HRD positive status ovarian cancer patients who are Platinum Sensitive Relapsed and have previously progressed on a licensed PARPi. 10. **Module 3:** Advanced recurrent or metastatic non-small cell lung cancer (NSCLC), or head and neck squamous cell carcinoma (H&N SCC). 11. **Module 4:** Any advanced solid tumours except gastric, gastro-oesophageal, oesophageal, or colorectal cancer with a small bowel resection. Ability to comply with an overnight fast of at least 10 hours prior to dosing and 4 hours after dosing as mandated, and ability to eat a high fat meal as mandated. 12. **Module 5 All:** Ovarian fallopian tube or primary peritoneal cancer, previous treatment with PARP inhibitor, platinum-sensitive relapsed ovarian cancer. 13. **Module 5 Part B:** Known or suspected BRCA mutation, PALB2 mutation, RAD51C/D mutation or HRD positive status.

4.2 Exclusion Criteria 1. A diagnosis of ataxia telangiectasia. 2. Prior exposure to an ATR inhibitor. 3. Bad reaction to ceralasertib. 4. Unresolved toxicities from prior therapy or recent treatment with cytotoxic, corticosteroid, or any investigational medicinal product. 5. **Module 2:** Contra-indicated for treatment with olaparib. 6. **Module 3:** Contra-indicated for treatment with durvalumab. 7. **Module 4:** Mean resting corrected QT interval (QTc) > 470 msec or history of familial long QT syndrome. Patients with type I or type II diabetes. 8. **Module 5:** Known hypersensitivity to PARP including AZD5305.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Ascending doses of ceralasertib (e.g., 40 mg, 80 mg, or 160 mg given once or twice daily depending on the specific module and schedule) in

combination with module-specific agents: * **Module 1:** Carboplatin (ATC code: L01XA02; Trade Name: Paraplatin) * **Module 2:** Olaparib * **Module 3:** Durvalumab * **Module 4:** Investigates the effect of food on ceralasertib absorption and whether ceralasertib has an effect on QT parameters. * **Module 5:** AZD5305 * **Route of Administration:** Oral (ceralasertib, olaparib, AZD5305) and Intravenous (carboplatin, durvalumab). * **Duration of Treatment:** Modules operate in 21-day or 28-day cycles, continuing until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care medications per investigator judgment. * **Prohibited:** Concurrent anti-cancer therapy, investigational agents outside of the specified module combinations, and strong CYP3A inhibitors/inducers.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumour Imaging, Genomic Testing (if applicable)
Baseline	Day 1	First Dose of Investigational Regimen (ceralasertib +/- combination agent), PK/PD sampling
Interim	End of each Cycle	Safety Labs, Efficacy Assessment (RECIST v1.1 imaging), Toxicity Evaluation for DLTs
End of Study	Variable	End of Treatment Visit, Disease Progression Confirmation, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Safety & Tolerability:** The number of subjects experiencing adverse events (AEs) and serious adverse events (SAEs) including abnormal clinical observations, Dose-Limiting Toxicities (DLTs), abnormal Electrocardiogram (ECG) parameters, abnormal laboratory assessments, and abnormal vital signs that changed from baseline. * **Module 4 - Effect of food on ceralasertib absorption:** Intensive PK assessments after a single oral dose of ceralasertib (Part A) measuring Geometric mean and 90% CI for the ratio of fed:fasted in area under the plasma concentration-time curve from zero to the last measurable time point (AUC_{0-t}) and from zero to infinity (AUC). * **Module 4 - Effect of ceralasertib on ECG parameters:** Change from baseline in HR, PR, QRS, and QTcF (Δ HR, Δ PR, Δ QRS, and Δ QTcF). Categorical outliers for QTcF, HR, PR, and QRS. Frequency of treatment emergent T and U wave abnormalities. If a substantial HR effect is observed (absolute value of the largest LS mean Δ HR > 10bpm), other correction methods such as QTcI will be explored to remove HR dependence.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) per RECIST v1.1. * Progression-Free Survival (PFS). *

Pharmacokinetics: Maximum Observed Plasma Concentration (C_{max}), Time to observed C_{max} (T_{max}), and Area under the plasma concentration-time curve (AUC) for ceralasertib. *

Pharmacodynamic modulation (e.g., biomarkers of DNA damage).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Intensive safety monitoring for hematological toxicities (e.g., anemia, thrombocytopenia) and other AEs during the DLT evaluation period and throughout treatment.
 - **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor/regulatory authorities.
 - **Data Monitoring Committee (DMC):** Internal Safety Review Committee (SRC) utilizes a rolling-six design to safely guide dose-escalation decisions before opening expansion cohorts.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all patients receiving at least one dose) and Efficacy Evaluable Set (patients with measurable disease and post-baseline scans).
 - **Sample Size Justification:** Targeted enrollment of approximately 357 patients. A rolling-six design is utilized for the dose-escalation phase to accurately assess DLTs while maximizing recruitment efficiency.
 - **Handling of Missing Data:** Standard Phase 1 conventions; patients lacking sufficient safety follow-up during the DLT window may be replaced.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Regular onsite and remote monitoring of safety data, eCRF entries, and investigational product accountability.
 - **Audit Trail:** eCRF locking, query resolution via data clarification forms, and rigorous clinical data management protocols.
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11. Study Identifiers & Governance

- **Primary ID:** D5330C00004
- **Registry Number:** NCT02264678
- **Posting ID:** 415539917422220602
- **Sponsor/Lead Organization:** AstraZeneca

- **Study Title:** Ascending Doses of Ceralasertib in Combination With Chemotherapy and/or Novel Anti Cancer Agents
 - **Official Regulatory Title:** A Modular Phase I, Open-Label, Multicentre Study to Assess the Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumour Activity of Ceralasertib in Combination With Cytotoxic Chemotherapy and/or DNA Damage Repair/Novel Anti-cancer Agents in Patients With Advanced Solid Malignancies
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12. Clinical Framework (Oncology Specific)

- **Primary Condition:** Adv Solid Malig - H&N SCC, ATM Pro / Def NSCLC (Non-Small Cell Lung Carcinoma), Gastric, Breast and Ovarian Cancer
 - **Diagnosis Threshold:** Histologically or cytologically confirmed advanced malignancy not appropriate for standard treatment
 - **Medical Methodology:** Targeted DNA Damage Response (DDR) combination therapy
 - **Optimization Target:** Identification of the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D)
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13. Study Procedures & Methodology

- **Management Pathway:** Modular dose-escalation (Part A) followed by dose-expansion (Part B). Option to start further combination modules is based on SRC decision using emerging preclinical data and safety/tolerability information.
 - **Safety Protocol (HCP):** Rolling-six design safety evaluations and DLT observation windows.
 - **Biomarker Protocol:** Genomic screening for targeted expansions (e.g., ATM-deficient, BRCA-mutant, HRR-deficient status).
 - **Quality Audit Mechanism:** Independent radiological review and central safety laboratory assessments.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Variable, based on individual disease progression and survival tracking.
 - **Onsite Visit Cadence:** Day 1 of each treatment cycle (every 21 or 28 days), plus intensive PK/PD sampling days.
 - **Tumor Assessment Frequency:** Typically every 6 to 8 weeks per RECIST v1.1.
 - **Safety Review Interval:** Continuous throughout the DLT evaluation window and each subsequent treatment cycle.
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15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					